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PRM013	Surecan® Safety II non-coring needle	Date: 01/2026	Edition: 10

I. Risk management review

Product range	Surecan® Safety II, non-coring needles (SSII)
Applicable risks management plan	PRM013 - Risk Management Plan Surecan Safety II Ed14
Current product risk analysis	PRM013 - Analysis Surecan Safety II Ed 25
Intended use	Surecan® Safety II non-coring access port needle is designed to allow easy penetration into the subcutaneous access port to deliver fluids and/or for blood withdrawal when recurrent or mid to long-term fluids must be injected into the vascular system and/or blood sampling/transfusion is required frequently. Surecan® Safety II non-coring access port needle is sterilized using ethylene oxide, is sterile, non-pyrogenic, in unopened, undamaged, individual packaging. Surecan® Safety II non-coring access port needle may be used for Contrast Enhanced Computerised Tomography (CECT) using high pressure injection. For power-injection of contrast media, the maximum recommended pressure is 325 psi (22.4 bar).
Benefits	The direct benefit link to the Surecan® Safety II non-coring access port needles for the patient is : - to receive his treatments during 7 days and avoid multiple needle punctures during this period of time

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II. Evaluation of overall residual risk acceptability

II.1 Check of the hazard/hazardous situations exhaustivity

Based on the following inputs, all known and foreseeable hazard/hazardous situations all been identified according to the intended use and considering all the lifecycle steps. Reasonably foreseeable misuse and the characteristics related to safety in both norma and fault conditions have been envisaged.
Inputs used:

- Design input : Product description, intended use, Standards to be applied (standard listing nr 006), Design Matrix (MMC), Device validation report (DVR006), IFU of similar products, Use specification SSII, Usability file (USR013); biological risk assessment (BRA006), Risk analysis of the Caresite ®.
- Clinical input : State of the Art and Clinical evaluation report (CER016),
- Manufacturing input: Process description nr 006, Process FMEA (SSII assembly process FMEA(HC-MY01-RA-N-PC-034), Multivac R245 S1 & S3 Packing FMEA(HC-MY01-RA-N-PC-042), cleanliness validation (CVR006), process validation, biological risk assessment (BRA006), sterilization evaluation report (SER004).
- PMS inputs: Complaints/vigilance cases for similar MDD products, market feedback, MAUDE data.

II.2 Check of completeness of risk control

All the risks have been reduced as far as possible. All decided risk control measures have been implemented except the addition of a clamshell around the needle inside packaging, which is currently in progress (PR0165). The implementation date is planned for Q1-2027.

Design risk control measures:
The implementation of the design risk control measures has been checked and are documented in the in MMC006 and in product specifications. The effectiveness of these measures has been proved thanks to verification tests which are gathered in the DVR006. In addition, the tests carried out with the users and documented in the Usability file USR013 as well as the design validation documented in the DVS006 have allowed to verify the effectiveness of certain design measures.

Manufacturing risk control measures:
The implementation of the manufacturing risk control measures has been checked and are documented in the Process Description nr 006.
All manufacturing related causes which could lead to a critical harm are minimized either by process validation or by 100% inspection. The Process description nr006 gathered all the elements proofs.
The effectiveness of the process validation is documented in validation reports mentioned in the risk analysis. The qualifications of test methods as Bioburden, packaging inspection and labels inspection are documented in validation reports.

The risks inherent to the Caresite valve itself, designed and manufactured by B.Braun USA, are managed under the Caresite Straight risk analysis SOP-MD-2009548. All the risks of the Caresite are identified, estimated, mitigated, and assessed as acceptable in the Caresite Straight Risk Analysis Ed13 dated on 20/02/2024. This risk analysis is a part of the Surecan Safety II non-coring needle risk management file.

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Information for safety:
The presence of the information for safety listed in the risk analysis has been checked in the IFU and more generally in labelling.
The IFU nr15327950 is aligned with this product risk analysis.

II.3 Risks arising from risk control measures

None of the risk control measures put in place generate any additional hazardous situations than those already identified. When they generate a potential cause of hazard /hazardous situation, the link between the measure and this cause is made in the RAM column. The cause is treated like any other, mitigated as much as possible and the resulting residual risk is assessed.

II.4 Individual residual risk acceptability

II.4.1 Risk acceptance matrix of the PRM013 – Risk Analysis – ED25

The PRM013 - Risk Analysis for Surecan Safety II non coring needle– ED25 allowed us to detect, estimate and evaluate 196 risks. The matrix below distributes them by level of Clinical Risk and technical probability of occurrence (Otech).

Risk acceptance matrix

	Clinical Risk (CR)				
OTech	CR1	CR2	CR3	CR4	CR5
(5) Frequent >1/1000	0%	0%	0%	0%	0%
(4) Probable ≤1/1000	0%	0%	0%	0%	0%
(3) Occasional ≤1/10.000	0%	0%	8% (16)	15.5% (30)	0%
(2) Rare ≤1/100.000	0%	14% (27)	12.5% (24)	8% (16)	0%
(1) Improbable ≤1/1.000.000	0%	0%	22.5% (45)	13.5% (26)	6% (12)

II.4.2 Risk profile

All risks have been considered within the risk analysis. In total 196 residual risks have been assessed.
Generally, 43% hazardous situations (ie 84 risks=30+16+26+12) were rated as high Clinical Risks (i.e. CR4-CR5). This reflects the clinical field of application of devices for vascular access.
12 (6%) high Clinical Risks (i.e. CR5) have been observed in the risk analysis. These 12 Clinical Risks show improbable technical probability of occurrence (Otech < 1 ppm).

In total,

- All the 196 evaluated risks are deemed acceptable because they are in the green area accordingly to the criteria established in the PRM013- Risk Management Plan Surecan Safety II-Ed14.
- 16% (30) evaluated risks were above the threshold for acceptable risks as defined in the risk management plan and therefore are discussed below. These risks all correspond to the same hazardous situation: leakage/bleeding. The risks are newly in the red zone as there is an increase in the rate of leak complaints since 2024 increasing the Otech to the occasional level. They are therefore discussed below.

II.4.3 Benefit-risk analysis of "red" residual risk

Despite mitigation, there is one hazardous situation which generates 30 residual risks judged above the threshold for acceptable risks in the risk analysis and that could not be further reduced. These risks are listed and discussed in the following tables in order to assess their benefit-risk balance.

Hazardous situation	Phase	Risk#	Potential causes	Current risk control measures
12. Leakage / Bleeding	Design	12.1	Assemblies design incorrectly defined / Assemblies not resistant to high pressure (325psi)	Needle assemblies designed to allow safe injection without leak or burst: Bond strength of the different assemblies > 15 N during 15 s. Detachment force > 18N. No leakage under 1.5 bars (21.0 psi) according to JIS T3221 Burst pressure above 22.4 bars (325 psi)
		12.2	Insufficient assembly resistance of the Caresite	Caresite assembly designed to allow transport and product handling without disconnection. Unscrewing torque > 0,2N.m
		12.3	Connectors not suitable with devices (syringes, lines, 3-way stopcock …) on the market / not luer compliant	Connectors compatible with luer connection according to ISO 80369-7 requirements
		12.4	Luer connector cracked. Luer connector not sufficiently resistant	Unscrewing of the Luer connector is possible even if strongly connected. Screwing torque of 0.6N.m is 5 times higher than the one requested by the ISO 80369-7
		12.5	Materials poor ageing evaluation	Needles performances tested after 3 (with Caresite) and 5 years (without Caresite) of aging.
		12.6	Insufficient resistance to chemical attack	Choice of material resistant to use of drugs, chemotherapy, Compliance with chemical requirements of the ISO 8536-4
		12.7	Insufficient torque limit	Unscrewing of the Luer connectors possible even if strongly connected. Screwing torque > 0.6N.m (5 times higher than the one requested by the ISO 80369-7) Unscrewing torque of Caresite/Y-site assembly ≥ 0.2 N.m
		12.8	Poor pinch clamp efficiency	The pinch clamp is designed to guarantee no leakage even if there are 50 clamping actions during the needle use.
		12.9	Pinch clamp too aggressive	The pinch clamp is designed to guarantee no leakage evebn if there are 50 clamping actions without any damage of the tubing
	Manufacturing	12.10	Nonconformity of the luer connector (dimensions, surface, circularity, crack)	Supplier luer connector validation. Incoming inspection of the luer connectors: HC-MY01-M-5-4-10-384-0 Final inspection of the luer connectors: HC-MY01-M-5-4-10-602-0
		12.11	Raw material not compliant with the specification (wrong material)	Incoming inspection of components
		12.12	Pinch clamp too aggressive	Incoming inspection of pinch clamps: HC-MY01-M-5-4-10-227-0
		12.13	Pinch clamp not functional	Incoming inspection of pinch clamps: HC-MY01-M-5-4-10-227-0
		12.14	Faulty Caresite	Quality inspection of the Caresite BBUS TP-MD-2000624
		12.15	Pierced tubing	Flow rate test of the needles 100 % checked by automatic machine - MPD102 Quality of the needles: HC-MY01-A-5-4-17-059-0 HC-MY01-A-5-4-10-073-0 HC-MY01-M-5-4-10-602-0
		12.16	Poor assembly of the different components / Insufficient of glue or CYC	Dimensional incoming inspection of the components - MPD100 Raw material incoming inspection of the components - MPD100 Needle assemblied by an automatic machine. Process validation - MPD102 Control and cleaning the glue nozzle/valve dispenser. Flow rate test of the needles 100 % checked by automatic machine. Quality of the needles: HC-MY01-A-5-4-17-059-0 HC-MY01-A-5-4-10-073-0 HC-MY01-M-5-4-10-602-0
		12.17	Poor assembly of the Caresite (insufficient glue quantity, Insufficient UV Cure)	Needle assemblied by an automatic machine. Process validation - MPD102 Flow rate test of the needles 100 % checked by automatic machine. Quality of the needles: HC-MY01-A-5-4-17-059-0 HC-MY01-M-5-4-10-602-0

		12.18	Caresite cracks	Quality inspection of the Caresite BBUS TP-MD-2000624 Flow rate test of the needles 100 % checked by automatic machine - MPD102 Quality of the needles: HC-MY01-A-5-4-17-059-0 HC-MY01-A-5-4-10-073-0
	Use	12.19	Maximum pressure exceeded: * straight version with or without Caresite : 325 psi * Y-site version with Needle free valve: 45 psi through the principal line & 325 psi through the Caresite	IFU (§5 Do not exceed the recommended pressure (325 psi - 22.4 bar) and access port maximum flow rate recommended by the manufacturer, as device failure may occur.)
		12.20	Luer connector cracked through over tightening	IFU (§2-f The Surecan® Safety II non-coring access port needles are designed exclusively for healthcare practitioners who must have experience with the technical and clinical requirements of non-coring needle insertion through subcutaneous access ports, removal and complications.)
		12.21	Contrast media injected not warmed	IFU (§5 Contrast media should be warmed to 37°C (98.6°F) before use. Failure to do so will result in up to 50% lower flow rates and/or device failure.)
		12.22	Connection insufficiently tightened	IFU (§5 Tighten all connections or needleless connectors before use. Failure to attach an end closing cone after removing a male Luer locking end closing cone can result in an air embolism or bleeding.)
		12.23	Needle chosen too short	IFU (§5 Choose the needle length according to the port reservoir depth, tissue thickness, and the thickness of any dressing beneath the base of the needle. If it is too long, the needle and/or subcutaneous access port may be damaged during insertion. If it is too short, the needle may not completely pierce the port septum, and medication may be delivered into surrounding tissue and/or the needle may be blocked. When placed correctly, the wings of the needle lie horizontally against the skin.)
		12.24	Needle insufficiently inserted into the septum	IFU (§5 Choose the needle length according to the port reservoir depth, tissue thickness, and the thickness of any dressing beneath the base of the needle. If it is too long, the needle and/or subcutaneous access port may be damaged during insertion. If it is too short, the needle may not completely pierce the port septum, and medication may be delivered into surrounding tissue and/or the needle may be blocked. When placed correctly, the wings of the needle lie horizontally against the skin. / The length of the needle written on the labelling corresponds to the length between the bottom of the foam and the canula tip.)
		12.25	No adhesive dressing covering the needle / Needle not correctly fixed by an adhesive dressing	IFU (§6-a Dress and secure the site per the local protocol (if dressing is not transparent, daily inspection is recommended).
		12.26	Needle inserted into the Needle free valve	IFU (§2-f The Surecan® Safety II non-coring access port needles are designed exclusively for healthcare practitioners who must have experience with the technical and clinical requirements of non-coring needle insertion through subcutaneous access ports, removal and complications.)
		12.27	Traction, stresses applied on the needle by patient movements	IFU (§10 The following information must be explained to the patient and/or to the person responsible for the patient: The repercussions of the device on the patient's way of life (e.g. physical activities, sport, etc.).
		12.28	For SSII with Y-site: HP injection through the non-HP line	Pinch clamp on the non-HP line indicates 45 psi max. IFU (§5 - For needles with Y-site, verify which pathways are high-pressure resistant. Applying high pressure in tubing that is not resistant to high pressure could potentially cause leakage)
		12.29	Tubing accidentally damaged by a sharp object	IFU(§2f The Surecan® Safety II non-coring access port needles are designed exclusively for healthcare practitioners who must have experience with the technical and clinical requirements of non-coring needle insertion through subcutaneous access ports, removal and complications.)
		12.30	Needle not primed/flushed before use on patient, so leak not detected before use	IFU (§5 Flush the needle before use. For references with Y-site, the Caresite® must be flushed too. Flushing makes it possible to verify the needle patency and purge the air from it.)

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Benefit - Risk assessment:

The benefit of the Surecan safety II needle is to allow the patient to receive his treatments during 7 days and avoid multiple needle punctures during this period of time.
Leakage/bleeding on Surecan safety II needle during lifetime might lead to Drug extravasation, Skin damage, Needle replacement, Air entry, Bleeding. Catheter rupture can be caused by several design, manufacturing or use reasons. The maximum severity associated to these potential damages is considered as High.

The complaint rate for the risk of leakage is 16 ppm. This led to Otech = Occasional and then to a risk (SxOtech) judged Inacceptable according to the acceptance matrix of the PRM013- Risk Management Plan Surecan Safety II-Ed13.

The reason of this increase of the complaint rate is due to a leakage observed at the cannula-wing junction due to no glue at the bottom of the wing, on some batches manufactured in 2023 and first part of 2024.

For Assembly Machine #332, the root cause is the following:

No purging was performed upon the replacement of the gluing nozzle's needle due to inadequate instruction provided in the SOP Surecan Safety II Needles Assembly, HC-MY01-A-5-4-17-007-0.

For Assembly Machine #333, the root causes are the following:

1) Inadequacy of SOP Defect Catalogue – Surecan Safety II, HC-MY01-A-5-4-17-055-0 in which no complaint's defect was listed as rejection criteria. Hence, no checking on the complaint's defect was performed by the operator during 100% manual inspection.
2) No detail step by step process requirement of cleaning the glue nozzle/valve dispenser is provided in the SOP thus insufficient glue potentially occurs due to glue nozzle blockage.
3) Inefficiency of the glue dispense area as the glue nozzle was targeting the cannula instead of wing hole. The glue dispense area was determined by the position of the glue nozzle.
4) No control mechanism on the correct position of the glue nozzle after bracket adjustment. Control mechanism on the correct glue nozzle is required as the position of the glue nozzle is affecting the glue dispense area.

An actions plan has been implemented on each machine between February 2024 and October 2024. This has effectively reduced leaks related to the absence or insufficiency of glue at the cannula-wing junction: Since October 2024, that is to say 9 months, we have not received any new complaints for this reason. The correctives actions are considered effective.
The complaint rate for 2025 is still "Occasional" because the batches are not all consumed (shelf life 5 years) but we expect a decrease in this type of complaints in the coming months.

Regarding the severity of potential harm to the patient, it should be noted that this type of leak caused by a manufacturing defect can be easily detected when flushing the needle before insertion into the patient access port (as it is recommended in the IFU and in the nursing guidelines) and then, the defect therefore does not cause any harm to the patient.

The aforementioned information allows us to judge the 30 current residual risks linked to leakage of Surecan Safety II needle as acceptable when these are compared to the commonly recognized benefits of this type of devices.

II.5 Overall residual risk evaluation

All clinically relevant risks related to design, to manufacturing, to use and use error of Surecan® Safety II non-coring needle was identified, estimated, mitigated, and assessed in the PRM013 Risk Analysis Ed25.

- The described risk control measures are implemented, and their effectiveness has been verified. The Risk Analysis can, therefore, be regarded as complete and adequate.
- 196 risks have been treated. All were mitigated as far as possible. One additional mitigation measure is currently in progress to reduce the risk that the needle arrives damaged before use: cannula bent or ruptured in packaging or tubing tangled due to memory effect (. The sterile packaging will be improved: A clamshell protecting the needle and allowing a good positioning of the tubing without knotting will be placed around the needle in Q1-2027).
- The design of the Surecan® Safety II non-coring needle includes a safety mechanism which reduces the risk of accidental needle-sticks during the removal of the needle. Indeed, the current complaint rate documented in the state of the Art for needle stick injuries is between 0.16% and 2% which is higher compared to the current rates of similar risk for the Surecan® Safety II non-coring needle (0.0002%).
- 166/196 risks have been assessed as acceptable accordingly to the acceptance criteria of the Risk Management Plan. No further additional risk mitigation measures were deemed necessary.
- 30/196 risks were judged as not acceptable in accordance with the acceptance criteria of the Risk Management Plan. Consequently, a benefit-risk analysis of these individual residual risks, based on State of the Art, has been performed (§II.4.3). This analysis allowed to consider these 30 residual risks as acceptable. No further additional risk mitigation measures were deemed necessary.
- All the causes related to the use or user error are addressed by specific recommendations in the IFU.
- The comparison of the rates of potential complication between the state of the art (when available) and current complaint rates of Surecan® Safety II non-coring needle (Otech) is in favor of the Surecan® Safety II non-coring needle. Even if we consider the major clinical risk (CR5) which is infection, the rate is very much lower than the complication rate reported in the state of the Art (0.00002% versus 0,3%). From a general perspective, the global complaint rate of Surecan® Safety II non-coring needle is very low (0.0046%) and often without any serious injury for the patient or user.

In total, the overall risk is assessed as acceptable for Surecan® Safety II non-coring needle, operated according to the intended use.

The global benefit-risk assessment is developed in the Benefit-Risk Report BRR006 Ed 03.

III. Effects of the results of the Risk analysis on other process

III.1 Effects of the Results on the Risk Management Plan

No effects on the Risk Management plan - PRM013 - Plan for Surecan® Safety II, non-coring needles – Edition 13 have been identified.

III.2 Effects of the Results on the Benefit-risk assessment

The Benefit-risk report nr BRR006 for Surecan® Safety II, non-coring needles Ed03 is aligned with the current risk analysis PRM013 - Ed25.

Note: If changes needed, immediately inform the PMS department.

III.3 Effects of the Results on the CER

No effects on the CER016 have been identified.
CER0016 for Surecan® Safety II, non-coring needles Ed02 is aligned with the current risk analysis PRM013.

Note: If changes needed, immediately inform the MSA department.

III.4 Effects of the Results on the PMS plan

No effects on the PMS plan have been identified. PMS-P nr 006 Ed02 for Surecan® Safety II, non-coring needles.

Note: If changes needed, immediately inform the PMS department.

IV. Risk management activities

IV.1 Implementation of the risk management plan

The PRM013 - Risk Management Plan Surecan® Safety II Ed14 has been appropriately implemented during this process of risk management, according to the NF EN ISO 14971+A11:2021standard and to the following general safety and performance requirements:

The requirements to reduce risks as far as possible [GSPR 2 - MDR 2017/745 UE]	PRM013 – Risk Analysis SSII – Ed 25 part Risk control measures
The requirements to have and maintain a risk management system [GSPR 3 - MDR 2017/745 UE]	SP 4-2-07-000-4
The requirements to reduce risks, so that the residual individual risk as well as the overall residual risk are judged acceptable [GSPR 4 - MDR 2017/745 UE]	PRM013 – Risk Report SSII – Ed 10 §II-4 & II-5
The requirements to eliminate or reduce risks related to use error [GSPR 5 - MDR 2017/745 UE]	PRM013 – Risk Analysis SSII – Ed 25 part Risk control measures, Phase # Use/User error
The requirements to have acceptable risks and side effects when weighed against the benefits [GSPR 8 - MDR 2017/745 UE]	PRM013 – Risk Report SSII – Ed 10 §II-5 & BBR006 Ed02
The requirements to reduce risks as far as possible [GSPR 2 - MDR 2017/745 UE]	PRM013 – Risk Analysis SSII– Ed25 part Risk control measures PRM013 – Risk Report SSII – Ed10 §II-1, II-2 & II-5

IV.2 Activities allowing to obtain relevant production and post-production information

The processes allowing the collect of production and post-production information are in place:

CER (Clinical evaluation) PMCF (Post market clinical follow-up) SotA (State of the Art) SAE during study	SP 4-1-07-001 SP 4-1-05 SP 4-2-04 SP 5-1-07
Complaints/vigilance cases for similar products (MAUDE) Complaints/vigilance cases for actual products Trend reports PSUR Information from the market	SP 5-4-01 & SP 5-4-04 SP 4-1-05
Process FMEA Caresite Straight Risk Analysis	HO-DDDD-G-4-1-05-140 COP-WI-8000003; MD-2009548
Corrective & Preventive Actions Nonconformity in production Product and process changes	SP4-2-01; HO-DDDD-G-4-2-07-001742 Inst. 4-2-01-001; HC-MY01-G-4-2-03 SP5-1-16; HC-MY01-G-4-2-04-002-0
Usability	SP 5-1-8-000-1

Production and post-market surveillances are active. The flow of information of production and post-production is ensured to collect and analyze relevant safety information.

Document history

Version	Change description
10	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed25). Addition of 8 new risks linked to " No or impossible fixation of the needle with a dressing". A total of 196 risks are assessed.
09	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed24). Assessment benefit / risk for risks linked to Leakage/bleeding.
08	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed23). Change of the § 2.2 following P608710_CLIN_C1 (Intended purpose, Indications, Contraindications) Change of the § 2.2 following P608710_CLIN_C2 (Performances, benefits) §IV.2-Change of the reference of the SP clinical SP 4-1-07-001 §IV.2-Addition of the SP Vigilance : SP 5-4-04
07	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed22).
06	Review of the Risk management report due to update of the PRM013 Risk Management Plan (Ed11) and the PRM013 Risk Analysis (Ed21).
05	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed20).
04	Review of the Risk management report due to update of the PRM013 Risk Analysis (Ed19).

Signatures

Title	Name
QSM Associate - PMS & Risk Management [Author]	Christelle JUNG
Quality Management System & Regulatory Affairs Manager [Reviewer]	Charlotte HENNAULT
Clinical Development Manager [Reviewer]	Patrick RAUGEL

Title: PRM013_Report_Surecan safety II non-coring needle Initiator: Christelle _ Jung

This document is signed electronically in compliance with the B. Braun electronic signature policies and procedures by following persons:

UserName: Jung, Christelle (jungchfr)
Title: SA-FR02-M-Post-Market Surveillance Manager
Date: Friday, 06 February 2026, 16:24 W. Europe Daylight Time
Meaning: Document signed as Author
==========

UserName: Jung, Christelle (jungchfr)
Title: SA-FR02-M-Post-Market Surveillance Manager
Date: Friday, 06 February 2026, 16:25 W. Europe Daylight Time
Meaning: Approve Document
==========

UserName: Hennault, Charlotte (dalychfr)
Title: QA/RA Director
Date: Friday, 13 February 2026, 16:42 W. Europe Daylight Time
Meaning: Approve Document
==========

UserName: Raugel, Patrick (raugpa16)
Title: SA-FR02-M-RAQA Director
Date: Thursday, 12 March 2026, 17:48 W. Europe Daylight Time
Meaning: Approve Document
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